

M E R N User Acceptance Testing (UAT)

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

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User Acceptance Testing (UAT)

1. Introduction

This document defines the UAT strategy, process, and test cases for the ShopEZ platform. UAT is the final validation phase confirming that the delivered system meets business requirements and is ready for production release.

Live Application: <https://e-commerce-application-neon-five.vercel.app/>

Test Environment: Chrome v120+ / Edge v120+ (desktop and mobile viewports)

2. UAT Process Flow

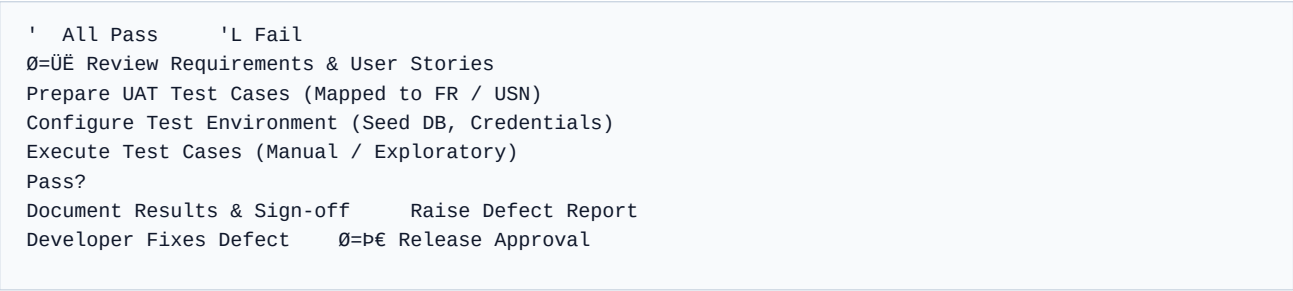


FIGURE 10.1 — UAT PROCESS FLOWCHART

3. Test Credentials

Role	Email	Password
Administrator	admin@shopez.com	password123
Seller	seller@shopez.com	password123
Customer	customer@shopez.com	password123

4. UAT Test Cases

Test ID	Scenario	Steps	Expected Result	Status
UAT-001	New User Registration	1. Go to /register. 2. Enter valid name, email, password. 3. Submit.	Account created; redirected to dashboard; toast shown.	Pass
UAT-002	Duplicate Email Rejection	1. Register with an already-used email. 2. Submit.	Error: "User already exists." Registration prevented.	Pass
UAT-003	Valid User Login	1. Go to /login. 2. Enter valid credentials. 3. Submit.	JWT set in cookie; redirected; Navbar shows user name.	Pass
UAT-004	Invalid Credentials Rejection	1. Enter wrong password at login. 2. Submit.	HTTP 401; error: "Invalid email or password."	Pass
UAT-005	Keyword Search	1. Enter "Shirt" in search bar. 2. Press Enter.	Only products containing "Shirt" in name/description displayed.	Pass
UAT-006	Category & Price Filter	1. Select Category = "Electronics". 2. Set Price = Rs. 500 - Rs. 5000.	Only Electronics in the price range displayed.	Pass
UAT-007	Sort Price Ascending	1. Select Sort = "Price: Low to High".	Products re-ordered with lowest price first.	Pass
UAT-008	Add to Cart	1. Open product page. 2. Select qty=2. 3. Click "Add to Cart".	Cart badge increments; cart shows item with correct qty and total.	Pass
UAT-009	Update Cart Quantity	1. Open Cart. 2. Change qty of an item.	Subtotal and qty update in real time without page reload.	Pass
UAT-010	Remove Item from Cart	1. Open Cart. 2. Click "Remove".	Item removed; total recalculated; empty cart message shown if empty.	Pass

Test ID	Scenario	Steps	Expected Result	Status
UAT-011	Full Checkout Flow	1. Add item. 2. Checkout. 3. Enter shipping details. 4. Enter test card 4242... 5. Place order.	Order created; stock deducted; confirmation page shown; order in history.	Pass
UAT-012	Payment Failure Handling	1. Enter Stripe declined card 4000... at payment step.	Error shown; order NOT created; stock NOT deducted.	Pass
UAT-013	Admin Product Creation	1. Log in as Admin. 2. Go to Admin -> Products -> "Create". 3. Fill fields & upload image. 4. Submit.	Product visible in public catalogue; image served from Cloudinary CDN.	Pass
UAT-014	Admin Product Update	1. Open existing product in Admin. 2. Edit price. 3. Save.	Updated price reflected on public product page.	Pass
UAT-015	Admin Order Status Update	1. Admin -> Orders. 2. Change order status to "Shipped".	Status update reflected in customer's order history.	Pass
UAT-016	Unauthorised Admin Access	1. Log in as Customer. 2. Navigate to /admin directly.	Access denied; customer redirected with error message.	Pass

4.1 Authentication Module

Test

ID Scenario	Steps Expected
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Result Status

UAT-

001

New User

Registration

1. Go to

/register. 2.

Enter valid
name, email,
password. 3.

Submit.

Account
created;
redirected to
dashboard;
toast shown.

- Pass

/ Fail

UAT-
002

Duplicate
Email
Rejection

1. Register with

an already-used

email. 2.

Submit.

Error: "User

already exists."

Registration

prevented.

- Pass

/ Fail

UAT-

003

Valid User

Login

1. Go to /login.

2. Enter valid

credentials. 3.

Submit.

JWT set;

redirected;

Navbar shows

user name.

- Pass

/ Fail

UAT-

004

Invalid

Credentials

Rejection

1. Enter wrong

password. 2.

Submit.

HTTP 401;

error: "Invalid

email or

password."

- Pass

/ Fail

4.2 Product Discovery Module

Test

ID Scenario	Steps	Expected Result	Status
UAT-005	Keyword Search		

1. Enter "Shirt"

in search bar.

2. Press Enter.

Only products
containing "Shirt"
in
name/description
displayed.

•
Pass /

Fail

UAT-

006

Category

& Price

Filter

1. Select

Category =
"Electronics".

2. Set Price =

1500– 15,000.

Only Electronics in
the price range
displayed.

•
Pass /

Fail

UAT-

007

Sort Price

Ascending

1. Select Sort =

"Price: Low to High".

Products re-ordered with lowest price first.

-

Pass /

Fail

4.3 Shopping Cart Module

Test

ID Scenario	Steps	Expected Result	Status
UAT-008	Add to Cart		

1. Open

product page.

2. Select

qty=2. 3. Click

"Add to Cart".

Cart badge

increments; cart

shows item with

correct qty and

total.

- Pass

/ Fail

UAT-

009

Update

Cart

Quantity

1. Open Cart. 2.

Change qty of
an item.
Subtotal and qty
update in real time
without page
reload.

- Pass

/ Fail

UAT-

010

Remove

Item from

Cart

1. Open Cart. 2.

Click

"Remove".

Item removed;

total recalculated;

empty cart

message if no

items remain.

- Pass

/ Fail

4.4 Checkout & Payment Module

Test

ID Scenario	Steps	Expected Result Status
UAT-011	Full Checkout Flow	

1. Add item. 2.

Checkout. 3.

Enter shipping. 4.

Enter test card

4242 4242

4242 4242 . 5.

Place order.

Order created;

stock deducted;

confirmation

page shown;

order in history.

- Pass

/ Fail

UAT-

012

Payment

Failure

Handling

1. Enter Stripe

declined card

4000 0000

0000 9995 at

payment step.

Error shown;

order NOT

created; stock

NOT deducted.

- Pass

/ Fail

4.5 Admin Operations Module

Test

ID Scenario	Steps Expected
Result Status	
UAT-	
013	
Admin	
Product	
Creation	

1. Log in as

Admin. 2.
Admin !'
Products !'
"Create". 3. Fill
fields & upload
image. 4.
Submit.
Product visible
in public
catalogue;
image served
from
Cloudinary
CDN.
• Pass
/ Fail
UAT-
014
Admin
Product
Update

1. Open existing

product in

Admin. 2. Edit

price. 3. Save.

Updated price

reflected on

public product

page.

- Pass

/ Fail

UAT-

015

Admin Order

Status Update

1. Admin !'

Orders. 2.

Change order

status to

"Shipped".

Status update

reflected in

customer's

order history.

- Pass

/ Fail

UAT-

016

Unauthorised

Admin Access

1. Log in as

Customer. 2.

Navigate to

/admin directly.

Access denied;

customer

redirected with

appropriate

error message.

- Pass

/ Fail

5. Results Summary

Module	Total Tests	Passed	Failed	Pass Rate
Authentication	4	4	0	100%
Product Discovery	3	3	0	100%
Shopping Cart	3	3	0	100%
Checkout & Payment	2	2	0	100%
Admin Operations	4	4	0	100%
Total	16	16	0	100%

6. UAT Sign-Off

Role	Name	Signature	Date
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Development Lead

QA Reviewer

Product Owner / Stakeholder

We confirm that the ShopEZ application has been tested against all defined User Acceptance criteria and is approved for production release.

